Double-Slit as Grid-Resolution Phenomenon in Grid-Rendering Cosmology

Working Note v0.2 (Revised)

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Abstract. Grid-Rendering Cosmology (GRC) treats quantum uncertainty as a grid-resolution phenomenon: $\Delta x \gtrsim l_g$, where l_g is the fundamental node scale of the discrete geometric substrate. This working note derives the consequences for the double-slit experiment within the GRC triad $\Psi_{\text{manifest}} = G(x) \otimes E(x,t) \cdot C(S(x,t))$. An exact UV-cutoff formulation is given, and the deviation from standard quantum mechanics (QM) is exactly defined and asymptotically evaluated for Gaussian slit profiles, showing exponential suppression: $|\Delta A| \sim \exp(-\pi^2\sigma^2/2l_g^2)$ for $\sigma \gg l_g$. The original polynomial ansatz $\mathcal{O}((l_g/d)^2)$ is hereby withdrawn. The primary empirically accessible distinction from standard QM is relocated to entropy-sensitive decoherence (P3) and higher-order measurement statistics (P4). The operational definition of local entropy S (gap G2) is identified as the critical open problem blocking empirical closure of P3.

Scope. Ontological and minimal-model extension of Grid-Rendering Cosmology to the sub-L0 quantum regime. This is not an empirical closure paper.

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1 Introduction and Motivation

The double-slit experiment is the canonical demonstration of quantum superposition and wave-particle duality [1]. Within standard QM, the interference pattern arises from the linear superposition of probability amplitudes propagating through both slits. The collapse of the pattern upon which-path measurement is attributed to entanglement with a measuring apparatus [4], but the ontological mechanism is left unspecified. Experimental demonstrations of wave-particle duality with massive particles [6] and the progressive character of decoherence [5] motivate a more explicit account of the transition from interference to localization.

Grid-Rendering Cosmology (GRC) provides an explicit ontological substrate: a discrete geometric grid G with node scale l_g, through which energy flows E and whose manifestation sharpness is modulated by the entropy-clarity function $C(S) = \exp(-S/S_{\max})$. Discrete spatial substrates have been explored in other quantum gravity frameworks [7]; GRC's specific claim is that the grid-resolution limit l_g canonically accounts for quantum uncertainty at the sub-Planck scale. Quantum uncertainty is thereby reinterpreted as a grid-resolution phenomenon: position cannot be realized below l_g. This working note asks whether this ontological identification gives rise to any testable physical distinction from standard QM.

This note is organized as follows. Section 2 states the axioms. Section 3 defines the double-slit setup in GRC language. Section 4 presents the minimal model. Section 5 states prediction requirements. Section 6 gives falsification points. Section 7 develops the UV-cutoff derivation and withdraws the polynomial ansatz. Section 8 states open gaps and priorities.

Status declaration. This is a working note, not a completed derivation. Results in Sections 2–6 are stable. The derivation in Section 7 is completed and its conclusion (exponential suppression) stands. Gap G2 (Section 8) is open.

2 Axioms

A1 — Grid (G). There exists a discrete geometric substrate G with a fundamental node scale l_g. The grid is not empty space, but the structural substrate on which spatial relations, neighbor-relations, and local topology are defined.

A2 — Energy Flow (E). Energy manifests as distributed flow through G. Localized energy concentrations appear as particle-like states; spatially distributed coupling appears as field-like states. E is a dynamic mode in G.

A3 — Clarity Function $C(S)$. Entropy S modulates manifestational clarity through

$$C(S) = \exp\left(-\frac{S}{S_{\max}}\right), \tag{1}$$

such that low S gives high clarity and high S gives low clarity. $C(S)$ determines the degree of spatial and dynamical sharpness in what can be registered.

A4 — Manifest Structure. Observable outcomes are always given by

$$\Psi_{\text{manifest}}(x,t) = G(x) \otimes E(x,t) \cdot C(S(x,t)). \tag{2}$$

No physical registration occurs independently of the triad G, E, and $C(S)$.

A5 — Grid-Resolution Limit. Spatial registration is limited by the grid's fundamental scale l_g. Position cannot be realized or measured with arbitrarily continuous precision, but only through grid-bound manifestations.

A6 — Registration as Local Selection. A measurement is a physical process that locally selects and stabilizes a manifestation of Ψ_{manifest} under given boundary conditions.

3 Definitions for the Double-Slit

D1 — Geometry as boundary condition on G. A double-slit setup is a physical boundary condition that selects a subset of allowed grid modes. Slit width d, slit separation D, and screen geometry determine which couplings between source and screen are geometrically accessible. G defines the total allowance field; slit geometry filters it.

D2 — Interference as distributed grid-coupling. When $C(S)$ is high, E maintains a distributed coupling over the allowed grid modes defined by the double-slit geometry. The registration distribution $P(x)$ on the screen is a weighted distribution over allowed positions. Interference is a geometrically responsive mode pattern in G under high $C(S)$.

D3 — Registration as local selection. A registration is a physical process in which a distributed manifest structure is selected and stabilized as a local event in a detector or on a screen.

D4 — Measurement as entropic localization. A which-path measurement introduces a local physical coupling that increases S at or near a slit. This lowers $C(S)$ locally, reduces distributed grid-coupling, and favors local mode realization over interference.

D5 — Scale limit. When relevant geometry approaches l_g, the continuous approximation loses validity. Standard interference predictions must then be corrected or replaced by an explicit grid-mode description. This constitutes a possible distinction from standard QM.

4 Minimal Model

M1 — Grid-mode basis. Let G define a discrete family of allowed spatial modes $\{\phi_n(x)\}$ over screen positions x, determined by slit geometry. The modes represent grid-bound registration patterns with resolution l_g. In the limit $l_g \to 0$, the continuous description of standard QM is recovered.

M2 — Energy coupling over allowed modes. Energy flow E realizes an amplitude distribution over $\{\phi_n(x)\}$ consistent with the boundary conditions. Under high $C(S)$, a distributed coupling is maintained over modes from both slits:

$$A(x) = \sum_n a_n \, \phi_n(x) \, C(S_n), \tag{3}$$

where a_n are mode coefficients determined by geometry and energy coupling, and S_n is the local entropic load associated with mode n.

M3 — Registration distribution. The screen registration distribution is

$$P(x) = |A(x)|^2. \tag{4}$$

In the double limit $C(S) \to 1$ and $l_g/d \to 0$, standard QM interference is recovered as the continuous approximation.

M4 — Decoherence via local S-increase. When a which-path measurement introduces local S-increase at one slit, the corresponding clarity factor for modes coupled to that path is reduced. The cross-terms in $|A(x)|^2$ are suppressed. In the limit of strong entropic injection, $P(x)$ approaches a single-slit-like pattern.

M5 — Leading GRC deviation (revised). *(This replaces the polynomial ansatz of v0.1.)* The UV-cutoff realization (Section 7) gives the exact deviation:

$$\Delta A(x) = \int_{|k|>\pi/l_g} \tilde{a}(k) \, e^{ikx} \, dk. \tag{5}$$

For Gaussian slit profiles with width σ, this is exponentially suppressed:

$$|\Delta A| \;\sim\; \exp\!\left(-\frac{\pi^2\sigma^2}{2l_g^2}\right), \quad \sigma \gg l_g. \tag{6}$$

The original leading-order polynomial ansatz $\mathcal{O}((l_g/d)^2)$ is hereby withdrawn.

5 Prediction Requirements

P1 — Backward compatibility (minimum requirement). GRC must recover standard QM in the double limit $C(S) \to 1$ and $l_g/d \to 0$. Failure here makes the model internally inconsistent. This is an analytic requirement, not an empirical prediction.

P2 — Geometric cutoff (principled but inaccessible). Under the UV-cutoff model, geometric deviations from standard QM are exponentially suppressed for $\sigma \gg l_g$. For sub-Planckian l_g, this deviation is undetectable with any foreseeable experiment. *P2 is downgraded from "primary test" to "principled but not practically accessible."*

P3 — S-sensitive decoherence component (primary candidate). Standard decoherence theory [2, 3] accounts for the loss of coherence via system-environment entanglement, with decoherence rates typically governed by temperature and the number and type of environmental couplings. The Lindblad master equation [8] provides the canonical framework. GRC predicts an additional modulation: decoherence is not determined only by temperature and known environmental coupling, but also by a local entropic load S that modulates clarity:

$$\tau_d \propto C(S) = \exp\left(-\frac{S}{S_{\max}}\right). \tag{7}$$

Systems with comparable temperature and conventional environmental parameters, but different local S, may exhibit different decoherence dynamics. This constitutes a testable departure from standard Lindblad evolution if and only if S can be defined operationally and independently of temperature (gap G2).

P4 — Gradual which-path suppression (secondary candidate). Partial local S-increase at one slit should produce gradual suppression of the interference term. Even when the mean pattern $P(x)$ overlaps with standard QM, GRC may be distinguishable through higher-order statistics: temporal correlation, event-time distribution, or shot-to-shot fluctuation structure under partial measurement.

P5 — No new prediction for $d \gg l_g$. GRC makes no deviation prediction in the regime where $d \gg l_g$ and S is low. The model must not produce undetectable explanations everywhere. The experimental window is narrow and specific.

6 Falsification Points

F1. The model fails if backward compatibility fails (analytic test; fails before any empirical test).

F2. The model fails if no scale-deviation exists where GRC predicts one. Given that P2 is now recognized as practically inaccessible for sub-Planckian l_g, F2 is a principled rather than operative falsifier under current technology.

F3. The model fails if decoherence is fully reduced to temperature and known environmental parameters, with no residual S-sensitive component. *This is the nearest-term empirical falsifier.*

F4. The model fails if higher-order statistics under partial measurement are identical to standard QM predictions across all strengths of induced S-increase.

F5. Internal consistency requires that the deviation $\Delta P(x)$ decrease with d/l_g in the UV-cutoff model. This is analytically supported by the exponential tail structure, but a full monotonicity claim for all geometries requires an explicit $P(x)$ derivation (gap G3).

F6 — What does not falsify the model. Standard QM results at $d \gg l_g$ do not falsify GRC. Failure to explain all QM phenomena does not falsify GRC. The model is only claimed where Section 5 asserts.

7 UV-Cutoff Derivation and Withdrawal of Polynomial Ansatz

7.1 Motivation

An earlier version of this note proposed a polynomial leading correction $\Delta P(x) \sim \mathcal{O}((l_g/d)^2)$ via Euler–Maclaurin applied to a discrete k-space sum. That approach failed on two grounds: (i) $\Delta k = 2\pi/l_g$ does not follow directly from a position-space grid spacing l_g; and (ii) for smooth, rapidly decaying $f(k)$, the Euler–Maclaurin correction integral vanishes. The polynomial ansatz is withdrawn.

7.2 UV-Cutoff Model

If l_g enters the double-slit problem as a minimum-length UV cutoff, the natural GRC amplitude is:

$$A_{\mathrm{GRC}}(x) = \int_{-k_{\max}}^{k_{\max}} \tilde{a}(k)\, e^{ikx}\, dk, \quad k_{\max} = \frac{\pi}{l_g}. \tag{8}$$

Standard QM uses the full Fourier integral without cutoff:

$$A_{\mathrm{QM}}(x) = \int_{-\infty}^{\infty} \tilde{a}(k)\, e^{ikx}\, dk. \tag{9}$$

The deviation is therefore exactly:

$$\Delta A(x) = A_{\mathrm{QM}}(x) - A_{\mathrm{GRC}}(x) = \int_{|k|>\pi/l_g} \tilde{a}(k)\, e^{ikx}\, dk. \tag{10}$$

This is not an ansatz. It is the exact consequence of the UV-cutoff model.

7.3 Evaluation for Gaussian Slit Profile

For a double-Gaussian slit with width σ and separation D:

$$\tilde{a}(k) = e^{-\sigma^2 k^2/2} \cos\left(\frac{kD}{2}\right). \tag{11}$$

The magnitude of the tail integral is bounded by:

$$|\Delta A(x)| \leq \int_{\pi/l_g}^{\infty} |\tilde{a}(k)|\, dk = \int_{\pi/l_g}^{\infty} e^{-\sigma^2 k^2/2}\, dk. \tag{12}$$

For $\sigma k_{\max} = \pi\sigma/l_g \gg 1$ (i.e., $\sigma \gg l_g$), this Gaussian tail is exponentially suppressed:

$$\int_{\pi/l_g}^{\infty} e^{-\sigma^2 k^2/2}\, dk \approx \frac{l_g}{\pi\sigma^2}\, e^{-\pi^2\sigma^2/(2l_g^2)}. \tag{13}$$

Therefore the deviation is bounded by:

$$|\Delta A| \lesssim \frac{l_g}{\pi\sigma^2} \exp\left(-\frac{\pi^2\sigma^2}{2l_g^2}\right), \tag{14}$$

and the geometric deviation from standard QM is thus exponentially suppressed in $(\sigma/l_g)^2$.

7.4 Conclusion

For any experimentally accessible slit width σ with sub-Planckian l_g, this ratio is not merely small — it is numerically indistinguishable from zero. The geometric distinction between GRC and standard QM via UV-cutoff is *principled but not experimentally accessible* with any foreseeable technology. P2 is downgraded accordingly.

8 Open Gaps and Next Targets

Table 1: Open gaps and their priority for v0.3.

Priority	Gap	Description	Blocks
1	G2	Operational definition of local S	P3 testability
2	P3	Formal decoherence model via $C(S)$	Main test candidate
3	P4	Concrete higher-order observable	Secondary test
4	G3	Explicit $P(x)$ for concrete geometry	Numerical check
5	G5	Closure Condition 3 vs. mode basis	Internal consistency

G2 (critical). The central open gap is an operational definition of local S that makes P3 testable without collapsing into a relabeling of standard environmental decoherence. Three candidates are under evaluation:

- **Candidate A** — von Neumann entropy of subsystem: $S_A = -\mathrm{Tr}(\rho \log \rho)$. Weakest; likely reduces to standard QM language.
- **Candidate B** — Entanglement entropy against environment. Requires showing $C(S_B)$ predicts a different rate or form than standard Lindblad evolution.
- **Candidate C** — Free-energy or structural complexity proxy. Most EFC-native; hardest to operationalize. Highest priority.

Criterion for G2 success. A definition of local S is only valid if it produces at least one of: (1) a distinct measurable proxy not reducible to temperature or known environmental couplings; (2) a different functional form; (3) a different dependence; or (4) a different statistical signature.

G3. An explicit $P(x)$ for a concrete slit geometry (Gaussian slits recommended) with mode coefficients a_n computed from boundary conditions and $C(S_n)$.

G5. Verification that the grid-mode basis $\{\phi_n(x)\}$ does not introduce hidden degrees of freedom beyond G, E, S, which would violate EFC Closure Condition 3 [11].

9 Status Summary

Table 2: Component status after v0.2.

Component	Status	Note
§1–§6 (axioms through falsification)	Stable	Canonical
M5 (geometric deviation)	Revised	Exponential, not polynomial
P2 (geometric test)	Downgraded	Principled, not accessible
P3 (S-sensitive decoherence)	Open	Primary test candidate
P4 (fluctuation statistics)	Open	Secondary test candidate
G2 (operational S)	Open	Highest priority gap

One-line status. The geometric cutoff test (P2) is in practice inaccessible as an empirical test for sub-Planckian l_g. The theory's quantum distinction, if any, lies in measurement as an entropy process (P3/P4), not in interference geometry.

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